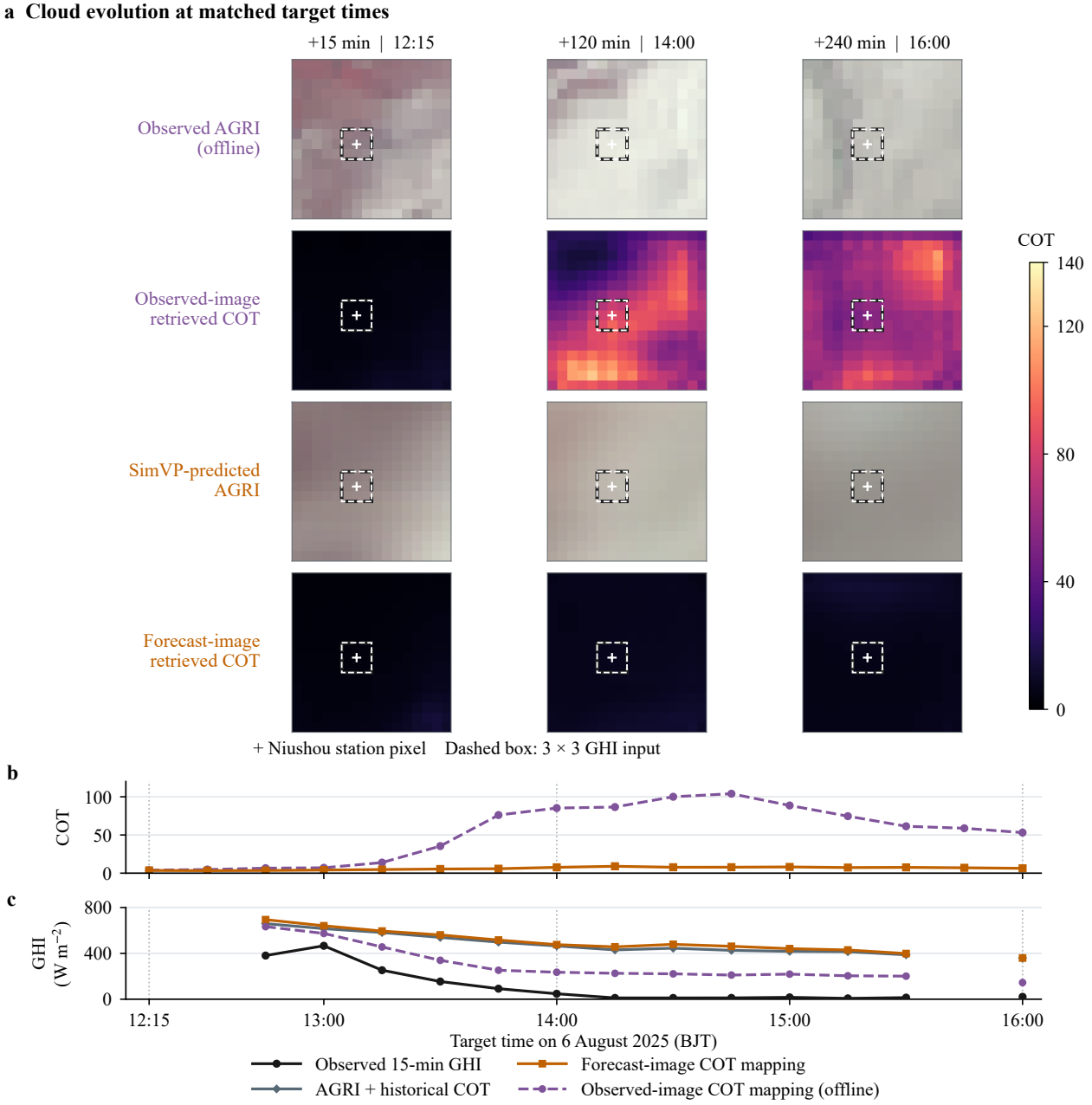

a Cloud evolution at matched target times
+15 min | 12:15
+120 min | 14:00
+240 min | 16:00
Observed AGRI (offline)
Observed-image retrieved COT
SimVP-predicted AGRI
Forecast-image retrieved COT
COT
140
120
80
40
0
+ Niushou station pixel
Dashed box: 3 × 3 GHI input
b
COT
100
50
0
c
GHI (W m−2)
800
400
0
12:15
13:00
14:00
15:00
16:00
Target time on 6 August 2025 (BJT)
Observed 15-min GHI
Forecast-image COT mapping
AGRI + historical COT
Observed-image COT mapping (offline)